

Artificial Intelligence in Pediatric Radiology

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CHLA Radiology Grand Rounds

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Objectives

- Describe the big players in radiology AI, both general and pediatric
- Summarize what radiology AI does well and what remains unresolved

Methods

Journals, Preprints: PubMed, Embase, arXiv, medRxiv

Conferences: NeurIPS, ICLR, ICML, CVPR, MICCAI, MIDL

Datasets: Hand-curated list

Newsletters: The Imaging Wire, RSNA News, TLDR AI, TLDR Tech, Signify Research, ESR / ECR, Radiology Business

Commercial: FDA list of AI-enabled devices

Date range: January 1 2008 – September 30 2026

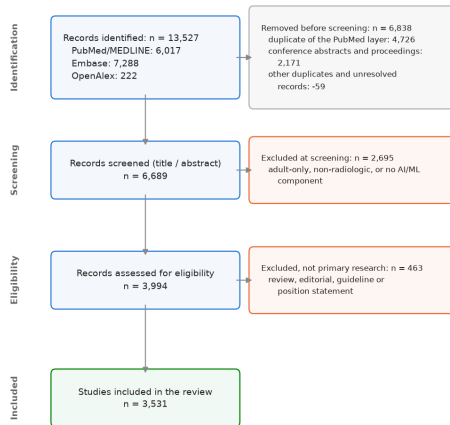
Search strategy: radiology AND AI [AND pediatric]

radiology: (radiology[tiab] OR radiological[tiab] OR radiograph*[tiab] OR "medical imaging"[tiab] OR "diagnostic imaging"[tiab] OR tomography[tiab] OR "magnetic resonance"[tiab] OR MRI[tiab] OR "MR imaging"[tiab] OR "MR images"[tiab] OR MRIs[tiab] OR "MR image"[tiab] OR "MR scan"[tiab] OR "MR scans"[tiab] OR "computed tomography"[tiab] OR CT[tiab] OR ultrasound[tiab] OR ultrasonograph*[tiab] OR sonograph*[tiab] OR echocardiograph*[tiab] OR mammograph*[tiab] OR "chest x-ray"[tiab] OR "x-ray"[tiab] OR fluoroscop*[tiab] OR scintigraph*[tiab] OR "positron emission"[tiab] OR PET[tiab] OR SPECT[tiab] OR angiograph*[tiab] OR neuroimaging[tiab] OR tractograph*[tiab] OR elastograph*[tiab] OR "fMRI"[tiab] OR "functional MRI"[tiab] OR "functional magnetic resonance"[tiab] OR connectome*[tiab] OR "diffusion tensor"[tiab] OR DTI[tiab] OR "bone age"[tiab] OR "skeletal age"[tiab] OR "skeletal maturity"[tiab] OR "barium enema"[tiab])

AI: ("artificial intelligence"[tiab] OR "machine learning"[tiab] OR "deep learning"[tiab] OR "convolutional neural network"[tiab] OR "convolutional neural networks"[tiab] OR "neural network"[tiab] OR "neural networks"[tiab] OR "computer-aided diagnosis"[tiab] OR "computer aided diagnosis"[tiab] OR "computer-aided detection"[tiab] OR radiomic*[tiab] OR "computer vision"[tiab] OR "large language model"[tiab] OR "large language models"[tiab] OR "foundation model"[tiab] OR "foundation models"[tiab] OR "transfer learning"[tiab] OR "vision transformer"[tiab] OR "self-supervised"[tiab] OR "random forest"[tiab] OR "support vector machine"[tiab] OR "gradient boosting"[tiab] OR "natural language processing"[tiab])

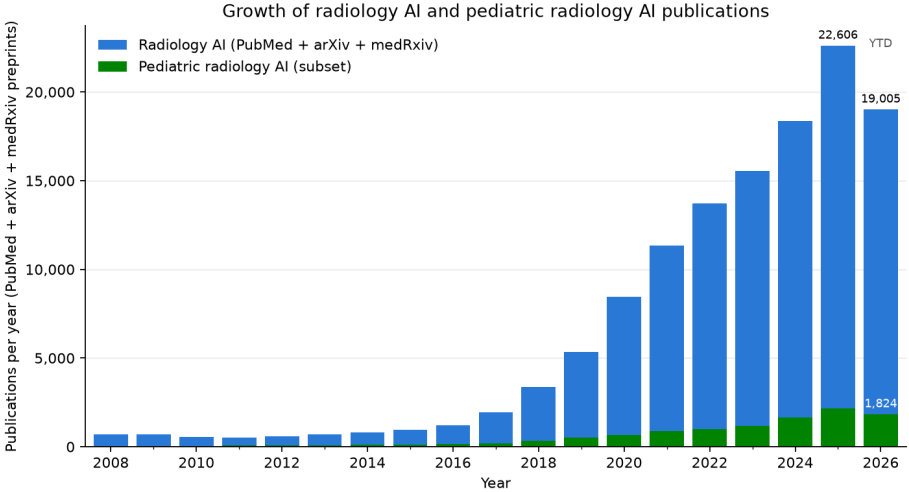
pediatric: (pediatric*[tiab] OR paediatric*[tiab] OR child*[tiab] OR infant*[tiab] OR neonat*[tiab] OR adolescen*[tiab] OR fetal[tiab] OR foetal[tiab] OR fetus*[tiab] OR foetus*[tiab] OR newborn*[tiab] OR preterm[tiab] OR "premature infant"[tiab] OR "premature infants"[tiab] OR "premature birth"[tiab] OR "premature neonate"[tiab] OR "premature neonates"[tiab] OR prenatal[tiab] OR antenatal[tiab] OR perinatal[tiab] OR youth[tiab] OR juvenile[tiab] OR "children's hospital"[tiab] OR schoolchild*[tiab] OR toddler*[tiab] OR baby[tiab] OR babies[tiab])

Study selection

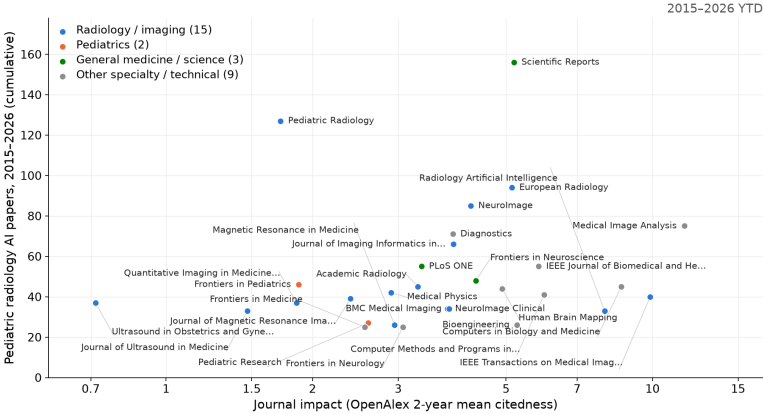


PubMed and Embase layers complete (2005-2026). Conference material is retrieved and reported but excluded by the publication-form criterion.

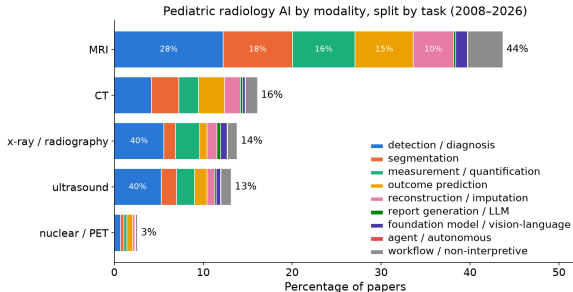
Radiology AI journal publication counts



Which journals publish pediatric radiology AI



Pediatric radiology AI publications, stratified by modality and task

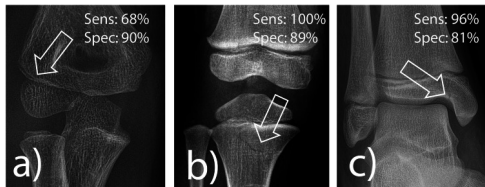


- **detection / diagnosis:** predict discrete disease labels (e.g., fracture, pneumonia, tumor, triage)
- **segmentation:** outline a structure or lesion
- **measurement / quantification:** measure a number from the image (e.g., bone age, Cobb angle, organ volume, fetal biometry)

- **outcome prediction:** a future risk, response, or survival estimate from the image (prognosis, radiomics signatures)
- **reconstruction / imputation:** generate a better or missing image
- **report generation / LLM:** text: draft, summarize, or extract from radiology reports

- **foundation model / vision-language:** a large pretrained model reused across tasks
- **agent / autonomous:** multi-step actions taken without a human in the loop
- **workflow / non-interpretive:** protocoling, scheduling, ordering, decision support, education

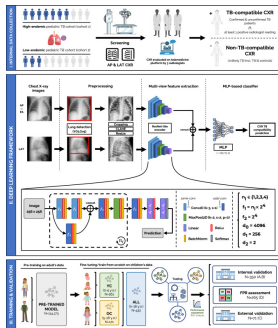
AI fracture detection validation of RBFrature (Radiobotics)



Ziegner et al., Eur Radiol 2025, figure 4: Frequently missed fracture entities with medicolegal significance according to van Laer. a Radial condyle fracture, (b) fracture of the proximal tibia, and (c) fracture of the medial malleolus.

source: doi.org/10.1007/s00330-025-11554-9

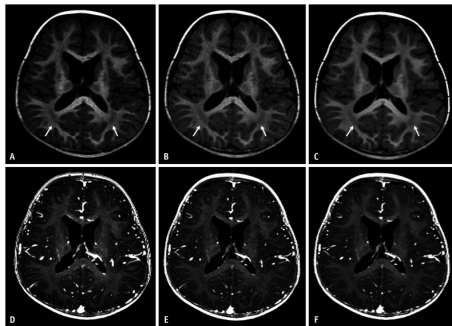
- **Problem:** pediatric fracture detection
- **Data:** 1,672 consecutive radiographs of children under 18 from one tertiary pediatric emergency department, plus a selected medicolegal set (proximal tibia, radial condyle); three pediatric residents read each before and after seeing the AI.
- **Method:** RBFrature (Radiobotics), a commercially available, CE-marked deep-learning fracture detector (adult-derived, pediatric extension) was run stand-alone and as a second reader.
- **Result:** stand-alone sensitivity 92%, radial condyle fracture sensitivity 68%, proximal tibia fracture sensitivity 100%, medial malleolus fracture sensitivity 96%. As second reader, AI raised resident sensitivity 84%→87% and accuracy 88%→90%; in 2% of cases readers dropped a correct diagnosis after seeing the AI.
- **Why it matters:** An external validation of a commercial pediatric fracture AI shows good performance, but also that even a high-performing AI can mislead readers in a small fraction of cases.



Capellan-Martin et al., Nat Commun 2025, figure 1: the multi-view pTBLightNet pipeline
source: doi.org/10.1038/s41467-025-64391-1

- **Problem:** pediatric pulmonary TB detection on chest radiographs
- **Data:** 114,173 adult chest radiographs for pre-training; 918 radiographs from three pediatric TB cohorts for fine-tuning and testing, including lateral views.
- **Method:** pTBLightNet, a deep-learning model using frontal and lateral views to detect TB-compatible findings; separate models for children under 5 and ages 5–18.
- **Result:** AUC 0.90 on internal testing and 0.68 on external testing; lateral views helped most in younger children. Age-specific models performed similarly to larger models trained across age groups.
- **Why it matters:** Adult pre-training can help with pediatric TB detection, but performance was substantially lower on external data.

Deep-learning reconstruction shortens pediatric brain MRI

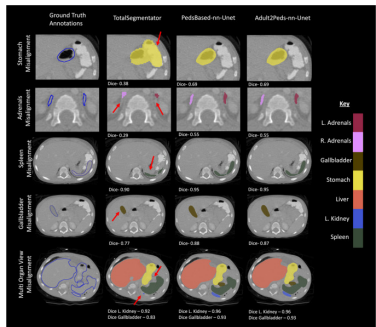


Yoo et al., Korean J Radiol 2025, figure 2: conventional (A, D) vs accelerated 3D T1 pediatric brain MRI without (B, E) and with (C, F) deep-learning reconstruction

source: doi.org/10.3348/kjr.2024.0701

- **Problem:** reducing pediatric brain MRI scan time while maintaining image quality
- **Data:** 46 children scanned with conventional and accelerated 3D T1 sequences, pre- and post-contrast, on one 3T scanner.
- **Method:** Vendor deep-learning reconstruction was applied to the accelerated scans; image quality measurements and radiologist ratings were compared with conventional scans.
- **Result:** Scan time decreased by 29% pre-contrast and 41% post-contrast. Deep-learning reconstruction gave lower noise, higher SNR and CNR, better overall image quality and fewer artifacts than conventional scans.
- **Why it matters:** Deep-learning reconstruction shortened 3D T1 acquisitions and improved image quality in this small, single-scanner study.

Validation of adult-trained TotalSegmentator on pediatric CT

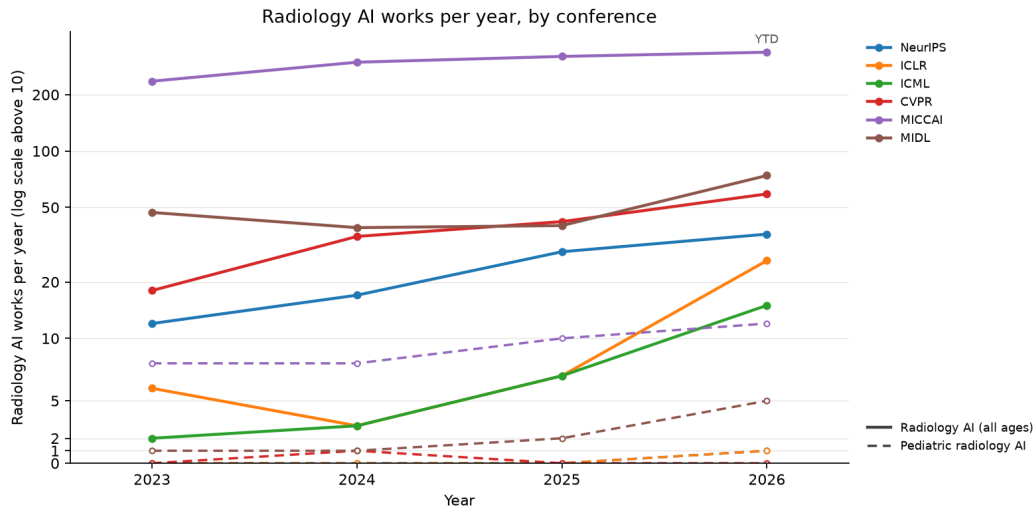


Chatterjee et al., J Imaging Inform Med 2025, figure 7: pediatric CT organ labels, ground truth vs adult-trained TotalSegmentator vs pediatric-trained models

source: doi.org/10.1007/s10278-024-01273-w

- **Problem:** abdominal organ segmentation on pediatric CT using an adult-trained model
- **Data:** 300 adult and 359 pediatric abdominal CT scans with expert organ labels for external testing.
- **Method:** TotalSegmentator, an open-source model trained on adults, was tested on both groups and compared with models trained or fine-tuned on pediatric scans.
- **Result:** Mean Dice score was 0.81 in adults and 0.73 in children, with the largest losses in the adrenals, duodenum and pancreas. Pediatric training or fine-tuning recovered the loss in performance.
- **Why it matters:** An adult-trained segmentation model performed worse in children; training on pediatric data improved performance.

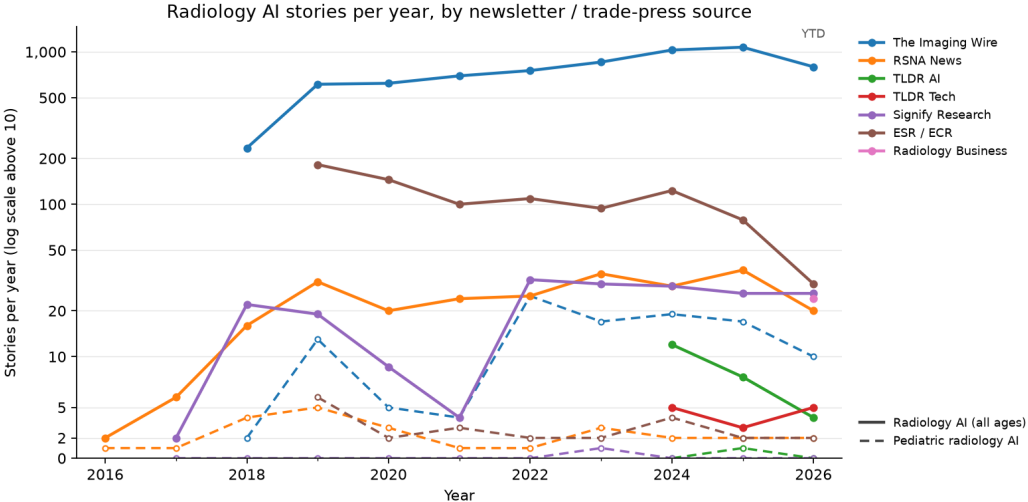
Radiology AI conference publication count



The biggest public pediatric radiology datasets

Dataset	Year	Modality	Size	Ages	Access
CBTN clinical MRI (Children's Brain Tumor Network)	2023	brain / spine tumor MRI	23,101 exams (1,526 patients); tumor masks for 370	pediatric	application (NCI CCDI)
GRAZPEDWRI-DX	2022	wrist radiographs (trauma)	20,327 images (6,091 patients)	0.2–19 y	open
RSNA Pediatric Bone Age Challenge	2017	hand radiographs	14,236 images (bone age + sex)	mean 10.6 y	open (RSNA terms)
FETAL_PLANES_DB	2020	fetal ultrasound (standard planes)	12,400 images (1,792 pregnancies)	fetal	open
ABCD Study	2018–	brain MRI, longitudinal	11,878 participants, up to 7 follow-ups	9–10 y at baseline	application (NBDC)
PediCXR / VinDr-PCXR	2023	chest radiographs	9,125 studies; 36 findings, 15 diagnoses	< 10 y	credentialed (PhysioNet)
Guangzhou pediatric chest X-ray (Kermany)	2018	chest radiographs	5,856 images (normal / bacterial / viral pneumonia)	1–5 y	open
ABIDE I + II (autism)	2012 / 2016	brain MRI (structural, resting fMRI)	2,156 subjects	5–64 y (mostly children)	open
Regensburg Pediatric Appendicitis	2023	abdominal ultrasound + labs	1,709 images (579 patients)	0–18 y	open
dHCP (Developing Human Connectome Project)	2022–24	neonatal + fetal brain MRI	886 neonatal + 297 fetal datasets, with segmentations	20–45 weeks	open (NDA)
Pediatric-CT-SEG (TCIA)	2022	chest / abdomen / pelvis CT	718 series (359 patients); 29 organ contours	5 days – 16 y	open
BraTS-PEDs (pediatric brain tumor challenge)	2023–25	brain tumor MRI	464 patients (2024 challenge); 1,649 series on TCIA	pediatric	registration / TCIA
FeTA (fetal brain tissue annotation)	2021–24	fetal brain MRI	280 volumes (120 train + 160 test); 7 tissue labels	18–35 weeks	registration

Radiology AI newsletter story count



Pediatric radiology AI in the trade press, 2026 (year to date) (14 stories)

Date	Newsletter	Story	Paper
2026-01-07	The Imaging Wire	BrightHeart Raises EUR 11M	
2026-01-28	The Imaging Wire	AHA Publishes New Stroke Guidelines	Prabhakaran et al., 2026 (Stroke)
2026-01-28	ESR / ECR	Guerbet: Innovation and French Cooperation in the Spotlight at ECR 2026	
2026-02-01	The Imaging Wire	DL-Based CT Reduces Radiation Dose	
2026-02-08	The Imaging Wire	GE Bolsters Ultrasound with Fetal AI	
2026-02-26	ESR / ECR	MRI Unveiling at the European Congress of Radiology (ECR) 2026	
2026-03-29	The Imaging Wire	Ultrasound AI Gets Breakthrough Nod	
2026-05-10	The Imaging Wire	Pediatric MRI Safety, RTs and Remote Radiology, and PSMA-PET	
2026-06-22	RSNA News	Researchers Use AI to Establish Body Charts for CT Imaging Across the Adult Lifespan	
2026-06-28	The Imaging Wire	BrightHeart's Bright Future	Zelop et al., 2025 (Obstetrics & Gynecology)
2026-07-12	The Imaging Wire	AI's Rise in Pediatric Imaging	Shah et al., 2026 (European Radiology)
2026-08-16	The Imaging Wire	CT vs. MRI for Pediatric TBI, Helium Vulnerability, and Ditching the Disk	
2026-08-25	RSNA News	AI Challenges Fuel Innovation Beyond the Leaderboard	
2026-09-23	The Imaging Wire	Repeat MRI Scans Take Operational Toll	

AI use in pediatric radiology departments

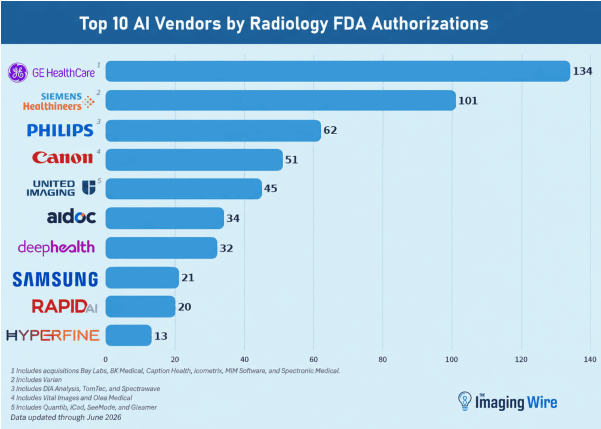
The Imaging Wire, 12 July 2026: “AI’s Rise in Pediatric Imaging” — Shah et al., *European Radiology* 2026 (doi 10.1007/s00330-026-12728-9)

- An international survey of **18 academic pediatric radiology centers** (69% response rate) asked which AI tools departments were using.
- **88.9%** reported using at least one AI tool. The most common uses were bone age (44.4%) and segmentation / quantification (22.2%). Only **16.7%** described AI as “transformational”.
- The main barriers were a lack of pediatric datasets (83.3%), PACS and workflow integration (66.7%), and cost (50%).
- Bone age is also the most common clinical application in our literature review and has an established commercial product.

Departments commonly reported using AI, but pediatric data and workflow integration remain barriers to broader use.

Radiology leads in FDA AI authorizations

The Imaging Wire, 14 September 2026: FDA AI-enabled device list, updated through June 2026



- **The FDA list:** the FDA's public list of AI-enabled devices authorized for the US market (510(k), De Novo, PMA). One row per authorization, scanners included; updated a few times a year, so it trails the newest decisions.
- **1,614** AI-enabled devices authorized since the FDA began tracking, up 5.9% from Q1 2026.
- **1,230** of them (76%) are radiology.
- The largest software-only vendors are Aidoc (34), DeepHealth (32) and RapidAI (20).
- Q2 2026: 89 authorizations, 66 (74%) radiology. Rules for generative AI in direct care are still pending.

Top 10 vendors by radiology FDA authorizations, counts include acquisitions

source: theimagingwire.com

A European cross-check: the Health AI Register

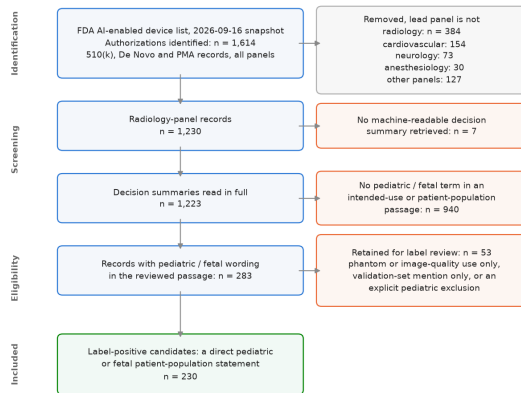
Health AI Register, radiology products (healthairegister.com), accessed 2 October 2026, matched against this deck's FDA radiology-panel list

- **What it is:** an independent, curated register of AI *software* for radiology that is CE-marked for the European market: **309 products from 121 companies**, each with its CE class, intended use and any FDA clearance. It leaves out scanners, which the FDA list includes.
- **Overlap:** 163 of the 309 (53%) also report FDA clearance. About 135 of those tie to a specific authorization in the FDA list (35 by submission number, the rest by company and product name). The other 146 are CE-marked only.
- **Coverage the other way:** the register's companies hold only about 40% of the 1,230 FDA radiology authorizations. Most of the rest are scanners and US- or Asia-only vendors.
- **Both lists lag:** BoneXpert was FDA-cleared on 22 September 2026 (K262390) and is not yet on the FDA list. The register still shows no US clearance for five products that the FDA list has (e.g., Gleamer BoneMetrics, Viz ANEURYSM).
- **Pediatric:** four of the 35 number-matched products carry a pediatric label in the FDA list (AZtrauma, MediAI-BA, EchoConfidence, 1CMR PRO). Several pediatric tools are CE-marked only: Gleamer BoneAge, AZboneage, Milvue TechCare Kids, Med-BoneAge.

Take-home: neither list is complete on its own. For a product of European origin, check both before concluding that it is or is not available.

How many cleared devices actually name children?

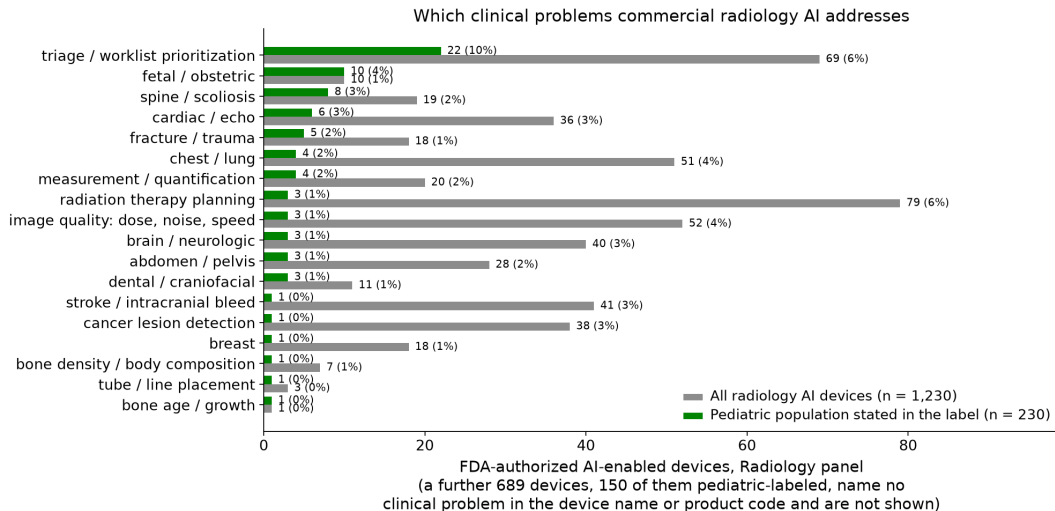
FDA AI-enabled device list: which authorizations name children



Keyword screen of the linked FDA decision summaries, not a determination of pediatric authorization: confirm the current labeling before purchase. One row per submission, so versions and resubmissions count separately.

230 of the 1,230 radiology-panel authorizations (19%) name a pediatric or fetal population in the labeling passage reviewed; 53 more name children only as a phantom, a validation set or an exclusion, and are kept for label review.

Which clinical problems the cleared products address



Products a pediatric radiologist should know (part 1)

Product	Vendor	Clinical problem	What it does	Pediatric scope	Where it stands
BoneXpert	Visiana	bone age / growth	Automated Greulich–Pyle and Tanner–Whitehouse bone age plus a Bone Health Index, from a hand radiograph	Built for children; CE-marked and in routine European use; FDA 510(k) cleared September 2026 [K262390, 2026]	Widely deployed (Europe)
BoneView	Gleamer	fracture / trauma	Fracture detection on trauma radiographs, presented as regions of interest alongside the image	US indication covers ages 2–21 for specified extremity views; pelvis, hip, femur, ribs and spine are adult-only [K222176, 2023]	Widely deployed
Rayvolve (AZtrauma)	AZmed	fracture / trauma	Fracture and MSK finding detection on radiographs; sibling modules cover chest findings, measurements and bone age	US fracture indication from age 2; the bone-age module is CE-marked only [K240845, 2024]	Widely deployed
TechCare Trauma	Milvue	fracture / trauma	Trauma radiograph triage: fractures, effusions and dislocations flagged for the reader	The decision summary states a pediatric population [K242171, 2025]	Early US deployments
EFAI Bonesuite XR Bone Age Pro	Ever Fortune.AI	bone age / growth	Greulich–Pyle bone age from a PA left hand and wrist radiograph, as an adjunct to the radiologist	FDA-cleared for ages 2–16 — the pediatric-specific US bone-age clearance [K234042, 2024]	FDA-cleared, early adoption
Physis	16 Bit	bone age / growth	Bone-age estimation with Greulich–Pyle atlas matching and draft report text	Pediatric tool; Health Canada licensed, no FDA authorization identified	Promising, limited US access
Fetal EchoScan	BrightHeart	fetal / obstetric	Flags eight morphological findings suspicious for congenital heart disease on second-trimester views	Prenatal by definition: second-trimester exams, maternal age 18+ [K252294, 2025]	Early deployments / pilots
Sonio Detect / Sonio Suspect	Sonio	fetal / obstetric	Checks that the required fetal views were obtained and are of usable quality (Detect); flags images suspicious for anomalies (Suspect)	Prenatal by definition; every FDA entry states a fetal population [K252433, 2026]	Early deployments / pilots
Delivery Date AI	Ultrasound AI	fetal / obstetric	Predicts the delivery date from a second- or third-trimester ultrasound when dating is unreliable	Prenatal; the first De Novo of its kind (a new device class) [DEN250007, 2026]	Newly authorized, unproven in practice
Swoop Portable MR + BrainInsight	Hyperfine	brain / neurologic	Bedside low-field brain MRI whose usable image quality comes from deep-learning reconstruction; BrainInsight adds automated volumes and midline shift	Decision summaries state a pediatric population; the natural use is the NICU and PICU, where transport is the risk [K253489, 2025]	In pediatric pilots
spineEOS / VEA Align	EOS imaging (Alphatec)	spine / scoliosis	Automated 3D spine and lower-limb modeling from biplanar low-dose full-body radiographs	Scoliosis follow-up is repeated imaging of a growing child, so the dose saving compounds [K251747, 2025]	Widely deployed (pediatric orthopedics)
CoLumbo	Smart Soft Healthcare	spine / scoliosis	Spine MRI segmentation, measurement and structured reporting, lumbar and cervical	Decision summaries state a pediatric population; no pediatric-specific performance reported [K254015, 2026]	Early deployments

Square brackets give the FDA submission supporting the pediatric claim and its decision year; a blank means no US authorization was identified for that product. “Where it stands” is our reading of deployment, not a measured adoption rate. A company’s total of FDA AI entries is not a pediatric credential — the large vendors’ entries are mostly whole scanners whose labels list pediatric imaging as one clinical application (see the authorizations-per-company figure).

Products a pediatric radiologist should know (part 2)

Product	Vendor	Clinical problem	What it does	Pediatric scope	Where it stands
BriefCase-Triage (CARE modules)	Aidoc	triage / worklist prioritization	Worklist triage on CT and CTA: intracranial hemorrhage, pulmonary embolism, pneumothorax, cervical-spine and rib fractures, free gas and more	Most entries state a pediatric population; the evidence behind them is adult, so treat pediatric use as local validation work [K253578, 2026]	Widely deployed
Viz LVO / Viz ICH	Viz.ai	stroke / intracranial bleed	Detects large-vessel occlusion and intracranial hemorrhage and pushes the case to the stroke team's phones	No pediatric population statement found in the screened summaries; pediatric stroke use would be off-label	Widely deployed (adult)
qXR-TB	Qure.ai	chest / lung	Chest radiograph screening for tuberculosis, built for high-burden settings with few readers	Vendor reports CE-marked use from age 0–15; US eligibility varies by module	Deployed in global-health screening
AIR Recon DL / TrueFidelity	GE HealthCare	image quality: dose, noise, speed	Deep-learning reconstruction: shorter MRI acquisitions at the same quality, and CT images at lower dose	Sold across ages; the pediatric benefit is fewer sedations and less dose, and it is the AI most children actually meet	Widely deployed
Deep Resolve	Siemens Healthineers	image quality: dose, noise, speed	MRI deep-learning denoising and resolution enhancement that supports accelerated acquisition	Vendor publishes pediatric protocols; scan-time reduction is the pediatric argument	Widely deployed
AIce	Canon Medical	image quality: dose, noise, speed	Deep-learning reconstruction for both CT (noise, dose) and MRI (noise, resolution, scan time)	Vendor publishes pediatric CT galleries; protocol and version determine what is available	Widely deployed
SmartSpeed / Precise Image	Philips	image quality: dose, noise, speed	MRI acceleration with deep-learning reconstruction (SmartSpeed); CT reconstruction at reduced dose (Precise Image)	Sold across ages; pediatric protocols are the vendor's own, not a separate indication	Widely deployed
SubtleMR / SwiftMR	Subtle Medical / AIRS Medical	image quality: dose, noise, speed	Vendor-neutral MRI post-processing: denoising and sharpening so a shorter acquisition still reads	No adult-only restriction in the reviewed US indications; SwiftMR's 2026 validation includes ages 0–21	Deployed, vendor-neutral retrofit
Butterfly iQ3; Clarius Bladder AI	Butterfly Network / Clarius	point-of-care ultrasound	Handheld ultrasound with on-device AI: bladder volume, gestational age, ejection fraction, lung and nerve tools	The scanners are labeled for adult and pediatric imaging, but each AI tool carries its own age scope and most are adult-only; Clarius Bladder AI covers ages 2 and over [K232257, 2023]	Scanners widely deployed; AI tools mostly adult-scoped
Arterys Cardio DL	Arterys (Tempus)	cardiac / echo	Cardiac MRI segmentation, flow and volume quantification with a cloud workflow	One of the earliest AI clearances to name neonates, infants, children and adolescents outright (2017) [K163253, 2017]	Historically important; now a Tempus product
Acorn 3D	Mighty Oak Medical	spine / scoliosis	Automated vertebral segmentation from CT to build patient-specific 3D models and surgical guides	Decision summaries state a pediatric population; the use case is pediatric deformity surgery [K260322, 2026]	Niche, surgical
Second Opinion Pediatric	Pearl	dental / craniofacial	Detects caries and other findings on pediatric dental radiographs	The rare device whose name and indication are pediatric from the start [K243893, 2025]	Deployed in dentistry

Square brackets give the FDA submission supporting the pediatric claim and its decision year; a blank means no US authorization was identified for that product. “Where it stands” is our reading of deployment, not a measured adoption rate. A company's total of FDA AI entries is not a pediatric credential — the large vendors' entries are mostly whole scanners whose labels list pediatric imaging as one clinical application (see the authorizations-per-company figure).

Product 1: BoneXpert — the most-validated pediatric AI you cannot buy

Visiana, Denmark (bonexpert.com) — bone age / growth

- **What it does:** automated Greulich–Pyle and Tanner–Whitehouse bone age from a single hand radiograph, plus a **Bone Health Index** from cortical thickness and an adult-height predictor. Not a second reader — it replaces the manual rating.
- **Why it is the benchmark:** designed for children from the start, in routine European clinical use since 2009, and the comparator nearly every academic bone-age paper measures itself against. It is the one pediatric AI with a decade of unglamorous production experience behind it.
- **The catch:** CE-marked in Europe; in the United States the vendor labels it **research use only**. There is no FDA authorization. The only pediatric-specific US bone-age clearance is a different product, EFAI Bonesuite XR Bone Age Pro (ages 2–16, K234042, 2024).
- **What it tells us about the market:** pediatric-native tools exist and work. The bottleneck is a US regulatory and commercial path, not the science.

Take-home: when someone says “there is no pediatric AI”, the accurate statement is that the best pediatric AI is not licensed here.

Product 2: Rayvolve — fracture detection with pediatric evidence

AZmed, France (azmed.co) — fracture / trauma; US pediatric fracture indication K240845 (2024)

- **What it does:** flags fractures on trauma radiographs; sibling modules add chest findings, orthopedic measurements (leg alignment, Cobb angle, hip angles) and bone age. **US pediatric clearance is fracture detection from age 2**; bone age is CE-marked only.
- **Independent pediatric validation:** 2,549 children in a pediatric ED (Dupuis et al., *Diagn Interv Imaging* 2022): sensitivity 95.7%, specificity 91.2% — but **significantly lower sensitivity under age 5 and in casted limbs**, the cases that matter most.
- **Head to head with BoneView** (Robin et al., *Pediatr Radiol* 2026, 366 children): AUC 0.87 vs 0.86; Rayvolve more specific (93% vs 87%), BoneView more sensitive (85% vs 81%); both helped junior residents. Neither is clearly better, but BoneView's US pediatric label excludes pelvis, hip, femur, ribs and spine.
- **The failure mode to watch:** in a pediatric ED, readers abandoned a correct diagnosis after seeing AI output in 2% of cases (Ziegner et al., *Eur Radiol* 2025).

Take-home: fracture detection is the most mature pediatric AI market, and still needs local validation by age band and anatomy, plus a policy for reader–AI disagreement.

Kamran review: broader research, stronger validation

Kamran et al., *Pediatric Radiology* 2026 (doi 10.1007/s00247-025-06462-5)

- **A large but concentrated literature:** 789 pediatric-focused original articles (2005–August 2024); 91% addressed image interpretation/diagnosis. Leading applications were bone age (88), pneumonia (65), and brain lesions/tumors (54).
- **Generalizability needs stronger evidence:** 65% relied on local hospital datasets. The authors call for diverse, multicenter datasets and prospective validation across pediatric ages, sites, and disease-prevalence settings.
- **Broaden the research agenda:** communication, education, policy, and stakeholder perspectives were underrepresented. Include children, families, and clinical teams in development and evaluation.

Take-home: prioritize representative data, validation, and neglected clinical needs. This review maps research activity; it does not compare model accuracy or establish clinical readiness.

ACR white paper: the pediatric AI gap

Sammer et al., *J Am Coll Radiol* 2023;20:730–737 (doi 10.1016/j.jacr.2023.06.003)

- **Scope:** ACR Pediatric AI Workgroup white paper framing limited access to pediatric AI as a health equity issue.
- **Availability:** at the time of the review, only 6 of more than 200 FDA-cleared tools in the ACR catalog indicated pediatric use (about 3%).
- **Why adult AI may fail:** growth, disease patterns, and imaging protocols differ; published studies show uneven performance across pediatric subgroups.
- **Barriers:** scarce labeled pediatric data, consent and privacy concerns, and weak financial incentives for smaller pediatric markets.
- **Recommendations:** validate in children, disclose age-specific evidence, fund pediatric development, involve families, and protect pediatric workflows. The authors introduce *Image IntelliGently* to advance this work.

Multisociety statement: safe pediatric AI implementation

Shelmerdine et al., *J Am Coll Radiol* 2026;23:89–101 (doi 10.1016/j.jacr.2025.08.019)

ACR, ESPR, SPR, SLARP, AOSPR, and SPIN propose four pillars for safe adoption:

- **Regulation and purchasing:** pediatric labeling and proposed safety ratings; diverse datasets, subgroup performance, transparency, and cybersecurity.
- **Implementation and integration:** define a local clinical need, select tools validated in children, pilot before scaling, and involve pediatric specialists and families.
- **Interpretation and postmarket surveillance:** retain human oversight; audit performance and patient outcomes; manage discrepancies, report adverse events, and address automation bias.
- **Education:** foundational AI literacy for all staff, pediatric-specific training, continuing education, and patient/public engagement.

Why is radiology AI particularly challenging in pediatrics?

- Small data
- Difficulty conducting clinical trials (FDA protection of children)
- Shortage of pediatric radiologists
- Lower perceived ROI
- Difficulty of generalizing from adult-trained models
 - Different and more variable body sizes compared with adults
 - Different diseases
 - Different imaging protocols

Most radiology AI is developed for adults

Safe for kids? AI medical devices in radiology overlook paediatric suitability

Kelly et al., European Radiology 2026

Intended Population	Devices	Age (%)
Adult Only	79	37
Paediatric Only	4	2
All Ages	24	11
Subsets of Paediatric Patients	18	8
Unclear	88	41
Total	213	100

Intended population of all 213 regulator-approved radiology AI devices in the Health AI Register (European approvals). Only 4 (2%) are labeled for children only; 41% do not state a population, and after checking literature and marketing, 41 of those 88 were still unclear.

Performance of adult-trained AI models in paediatric imaging: a scoping review

Laborie et al., European Radiology 2026

- 20 studies (2014–2025) ran adult-trained models, unmodified, on pediatric images: radiography, CT, MRI, DEXA, ultrasound
- 18 of 20 found worse performance in children
- Detection degraded most: pulmonary nodule sensitivity fell from 68–100% in adults to 26–68% in children
- Segmentation: Dice drops >0.10 across organs and modalities
- Children ≤ 2 years had the largest deficits for every task type
- Take-home: do not assume an adult tool works in children; validate locally, fine-tune, or retrain first

What the ACR says

<https://www.acr.org/Data-Science-and-Informatics/Informatics/pediatric-radiology-ai-resources>

- Children get left behind in AI-assisted triage
- Pediatric payments through CHIP are historically 30% lower than adult Medicare, which may disincentivize pediatric AI development
- There are mixed results when applying adult-trained AI to pediatric cases
- As of 2023, Of >200 AI Software cleared by the Food and Drug Administration (FDA) via the Software as a Medical Device (SAMD) pathway that are currently listed in the online ACR Data Science Institute (DSI) catalog, only six (3%) are intended for pediatric use
- See the website for videos about reimbursement, challenges, and developments

What the ACR says

Take home points



**Pediatric patients are
left behind in
radiology AI**



**There will not be
equitable AI for all
pediatric patients based
on \$ ROI alone**



**Even in pediatrics,
equitable is not one
size fits all**

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Worth knowing as a radiologist

- **Deep Learning: A Primer for Radiologists** (Radiographics paper, 2017): a helpful introduction to the principles and applications of deep learning in radiology; essential for understanding how AI tools work and their limitations
- **RSNA Pediatric Bone Age Challenge (2017)** (benchmark / dataset): a landmark pediatric benchmark; useful for understanding automated bone-age accuracy and its limits
- **Gleamer BoneView / AZmed Rayvolve** (clinical products): examples of fracture-detection assistance with pediatric indications; eligibility depends on age, anatomy, and product version
- **FDA AI-enabled device list and linked labeling** (regulatory resource): a starting point for checking authorized use and pediatric age ranges; the list is not exhaustive
- **Health AI Register** (product directory): commercial radiology AI products with CE / FDA status and intended use; confirm pediatric use in the labeling
- **CLAIM checklist (2024 update)** (reporting guideline): check whether data, reference standards, and testing are adequately described; reporting completeness alone does not establish clinical benefit
- **TotalSegmentator** (research tool): automated CT / MR organ segmentation for volumes and body composition; check accuracy in children before trusting measurements
- **nnU-Net** (research framework): a strong baseline for training segmentation models; ask whether a new method improves on an appropriate comparator
- **MedSAM / segment-anything models** (research models): prompt-guided contours can assist annotation; radiologists still need to review and correct the output
- **MONAI** (research framework): infrastructure for imaging-AI development; useful when collaborating with a research team, rather than a ready-to-use diagnostic tool
- **NVIDIA NV-Reason-CT** (open research model, Sept 2026): a 3D CT vision-language model that drafts chest / abdominal CT findings with step-by-step reasoning and answers questions; open weights, research use only, not FDA-cleared, and no pediatric data or age range reported

Explore our database: <https://pedrad-ai.vercel.app>

Pediatric Radiology AI Database



Articles, open source software, trade-press stories, commercial devices and public datasets for AI in pediatric radiology. Search, filter columns by value, sort, pick columns, and export any view as CSV.

Articles 3,496

Open source software 339

Newsletters 161

Commercial 997

Datasets 13

The FDA's list of AI-enabled medical devices, Radiology panel only (snapshot 2026-09-17; [source](#)), one row per device: a device the FDA authorized more than once (a new version or indication) is shown at its most recent authorization, with the earlier ones listed, so the 1,230 authorizations make 994 rows. The FDA describes the list as noncomprehensive. The 3 rows marked **not FDA-authorized** are pediatric products on our curated list that have no US authorization (for example BoneXpert, CE-marked and US research use only). **Pediatric use** is yes for the 172 authorized devices with at least one authorization whose linked label states a pediatric or fetal patient population or intended use (2026-09-16 screen); hover the yes for which. About half are whole scanners whose labels list pediatric imaging as one clinical application, so confirm the current authorization before purchase. **Curated pediatric product** marks the 23 products with a documented pediatric indication reviewed by hand on 2026-09-16; on an FDA row it is a company-level flag, not a claim about that row's authorization.

Search

Clear

Columns ▾

Export CSV

Examples: `pediatric:yes` `company:="GE HealthCare"` `date>=2025` `fetal OR pediatric` `us_status:="not FDA-authorized"` [show fields](#)

12 of 997 rows match.

Device	Company ▾	Decision date	Submission	US status ▾	Earlier authorizations	Product code ▾	Pediatric-use screen ▾	Pediatric use ▴ ▾
▶ EFAI Chestsuite XR Malpositioned ETT Assessment System (ETT-XR-100)	Ever FortuneAI	2025-02-20	K242821	FDA-authorized	—	QAS	label-positive candidate	yes
▶ EFAI Bonesuite XR Bone Age Pro Assessment System (BAP-XR-100)	Ever FortuneAI	2024-06-07	K234042	FDA-authorized	—	QIH	label-positive candidate	yes
▶ EFAI CARDIOSUITE CTA ACUTE AORTIC SYNDROME ASSESSMENT SYSTEM	Ever FortuneAI	2024-04-08	K240291	FDA-authorized	—	QAS	label-positive candidate	yes
▶ EFAI ERSUITE CT APPENDICITIS ASSESSMENT SYSTEM (APPEN-CT-100)	Ever FortuneAI	2026-04-22	K253163	FDA-authorized	—	QAS	—	no
▶ Seg Pro V3 (RT-300)	Ever FortuneAI	2026-01-28	K251306	FDA-authorized	—	QKB	—	no
▶ EFAI Neurosuite CT Midline Shift Assessment System (MLS-CT-100)	Ever FortuneAI	2024-12-06	K241923	FDA-authorized	—	QAS	—	no
▶ EFAI NeuroSuite CT ICH Assessment System	Ever FortuneAI	2023-10-04	K231025	FDA-authorized	—	QAS	—	no
▶ EFAI RTSUITE CT HCAP-Segmentation System	Ever FortuneAI	2023-09-25	K231928	FDA-authorized	—	QKB	—	no
▶ EFAI ChestSuite XR Pneumothorax Assessment	Ever FortuneAI	2022-11-08	K221552	FDA-authorized	—	QFM	—	no

Recommendations for pediatric radiologists

- AI development is constrained by scarce public datasets, fragmented institutional data, rare diseases, and heterogeneous reporting practices.
- Choose one task worth improving in your practice: bone-age measurements, fracture detection, or reducing CT dose / MRI scan time.
- Before adopting a tool, ask which children it was tested on. Check the intended age range and test it on your own cases, including younger children and unusual anatomy.
- In a local pilot, measure what changes with AI: reading time, missed findings, false alarms, dose, or scan time. Review errors and keep checking after rollout.
- Help build the evidence we need: choose a pediatric problem you see often, label cases carefully, and work with other hospitals to test whether a model works beyond the site that built it.